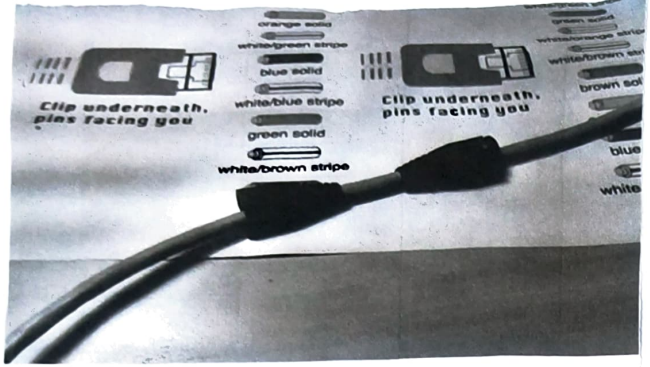
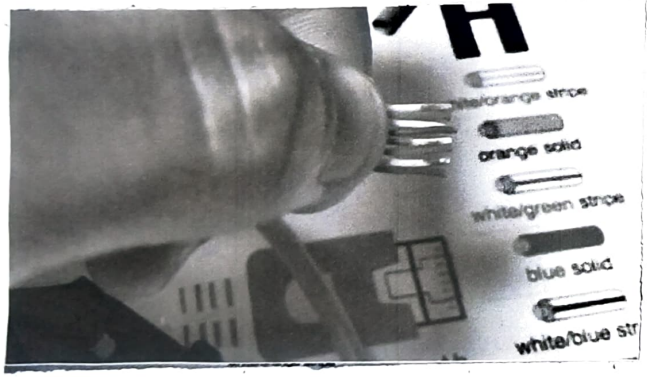


Step-1



Step-2



Step-3

Exno: 2 Date: 30.7.

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

* Aim:

To study the different types of network cables used in computer networking and to understand the construction and application of straight-through and cross-over Ethernet cables using RJ45 connectors and a crimping tool.

* Types of Network Cables

1. Unshielded Twisted Pair (UTP) Cable.

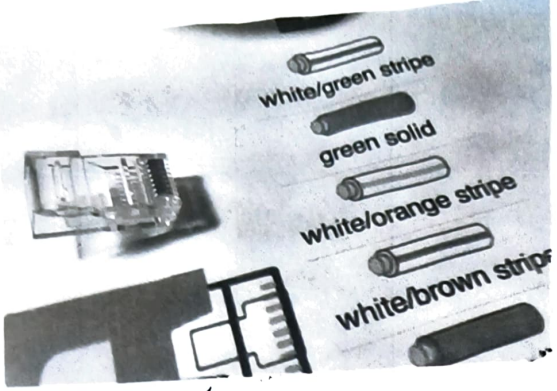
- Consists of four pairs of twisted copper wires.
- Common categories: Cat 3, Cat 5, Cat 5e.
- Supports speeds up to 1 Gbps (Cat 5e).

→ Advantages:

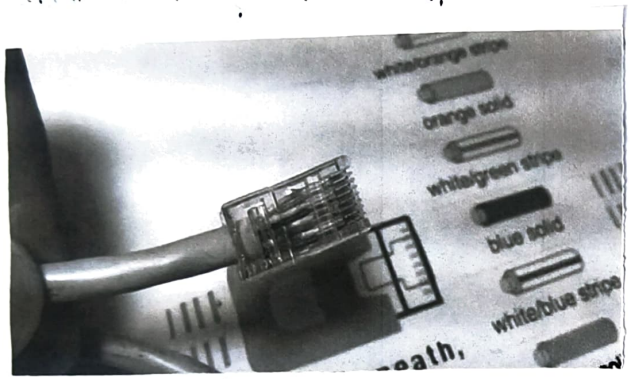
- Low cost
- Easy installation.

→ Disadvantages:

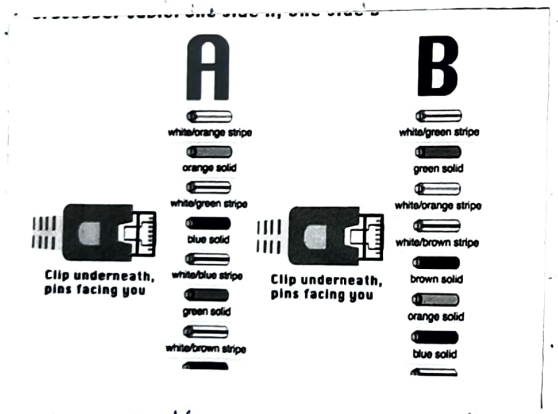
- Prone to electromagnetic interference (EMI).



Step 4



Step 5



Straight through network cable

→ Applications:

- Ethernet LANs
- Office and home networking.

2. Shielded Twisted Pair (STP) Cable

- Similar to UTP but includes protective shielding.
- Categories: Cat6, Cat6a, Cat7.
- Supports speeds up to 10 Gbps.

→ Advantages:

- Better noise immunity
- Higher speed.

→ Disadvantages:

- Expensive
- Difficult to install

→ Applications:

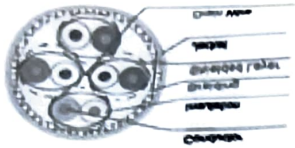
- Data Centres
- High-speed enterprise networks.

3. Coaxial Cable

- Contains a central copper conductor surrounded by insulation and shielding.
- Supports 10-100 Mbps.

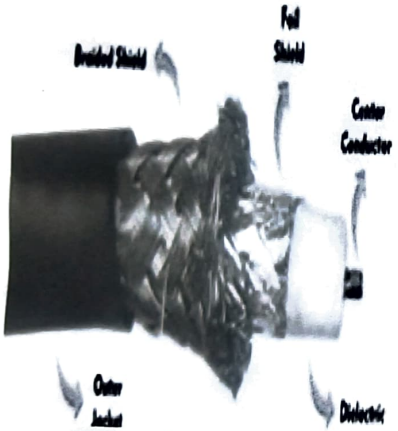
→ Advantages:

- High bandwidth.





SSTP



Coaxial cable

b) Straight-through Cable.

· A straight-through cable uses the same wiring standard (T568A-T568A or T568B-T568B) at both ends.

· => Used for Connecting:

· PC to Switch.

· PC to Router.

· Switch to Router.

=> Cross-Over Cable -

A cross-over cable uses T568A at one end and T568B at the other end.

=> Used for connecting:

· PC to PC

· Switch to Switch

· Hub to Hub

* Procedure -

1. Select the required length of UTP cable.
2. Strip approximately 1.5cm of the outer insulation using a cable stripper.
3. Seperate and untwist the four wire pairs carefully.
4. Arrange the coloured wires according to the required wiring standard (T568A or T568B).

5. Trim the wires evenly.

6. Insert the wires completely into the RJ45 connector.

7. Use the crimping tool to crimp the connector securely.

8. Repeat the same process for the other end of the cable.

9. For a Straight-Through cable, use the same wiring standard at both ends.

10. For a Cross-over cable, use T568A at one end and T568B at the other end.

11. Connect the cable between two networking devices.

12. Test the cable using a cable tester or by checking network connectivity.

* Result:

Thus, the different types of network cables and the construction of Straight-Through and Cross-Over Ethernet cables were studied successfully. and their connectivity was verified.